Electrical Engineering

Electronics and Communication Engineering

Digital Logic

Practice Sheet 01 Discussion
Notes

Combinational Circuits

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Question (MCQ) EASY

Which of the following is true?

A All gates follow commutative as well as associative law. ✗

B NOR & XNOR does not follow associative law. ✗

C NAND & XOR does not follow associative law ✗

D All gates follow commutative law but some gates doe not follow associative law. ✓

Question (MCQ) MODERATE

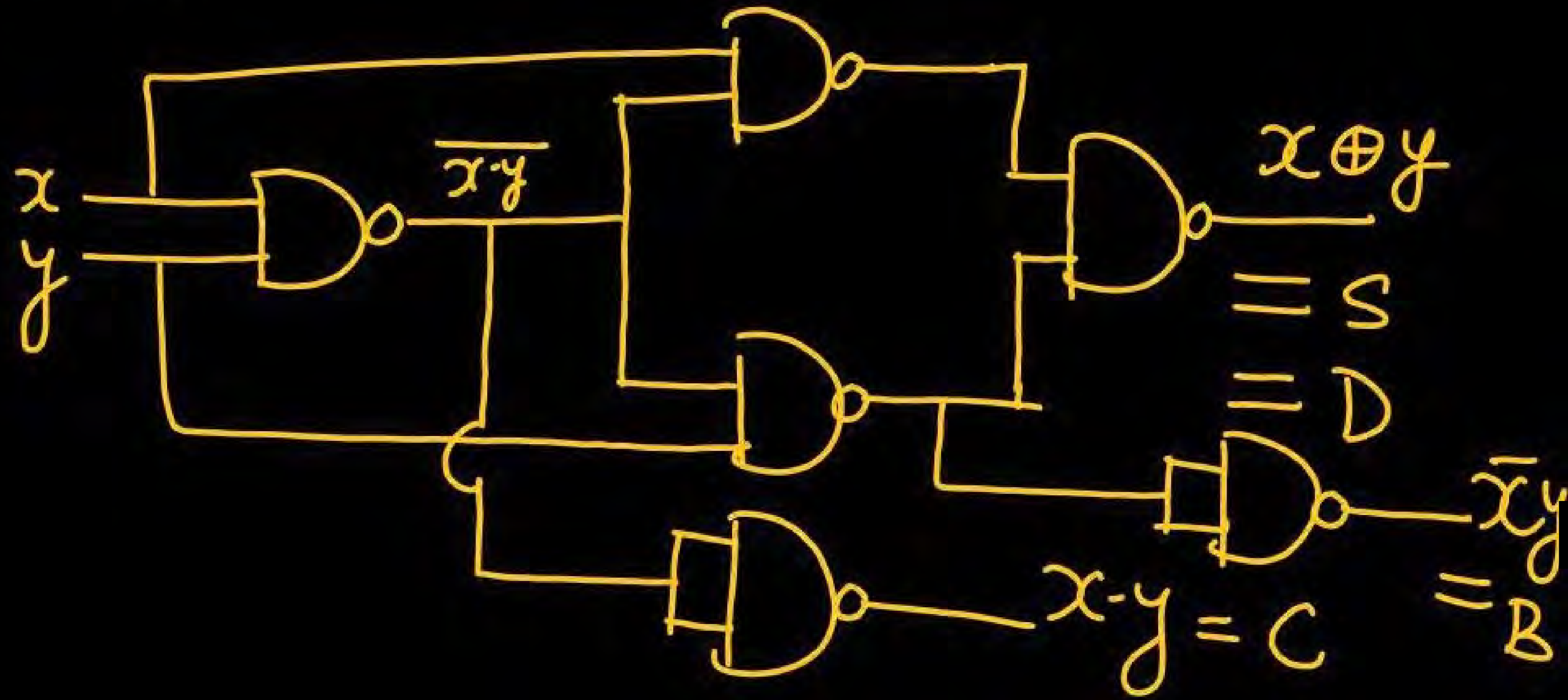
We have to implement half adder as well as half subtractor. Minimum number of 2-i/P NAND gate require is 6.

H.A. $\longrightarrow$ $S = x \oplus y$

$C = x \cdot y$

H.S $\longrightarrow$ $D = x \oplus y$

$B = \bar{x} y$



Question (MCQ) EASY

A digital circuit is as given below:

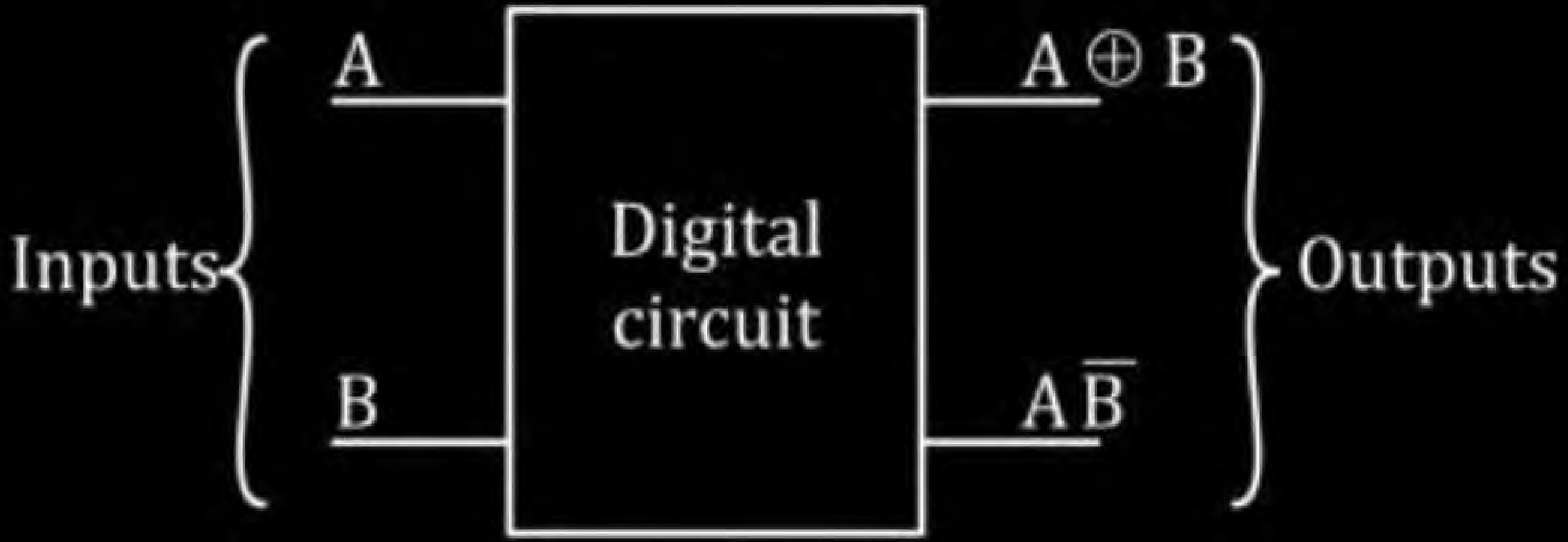
The circuit implements

A. $A + B$

B. $A - B$

C. $B - A$

D. None of these



$A - B \Rightarrow \text{Borrow} = \overline{A} \cdot B$

$B - A \Rightarrow \text{Borrow} = \overline{B} \cdot A = A\overline{B}$

Question (MCQ) MODERATE

$y_1' = (A\oplus B)\oplus \bar{A}B$

$= A\oplus B\oplus \bar{A}B = A\oplus[\overline{B}\cdot\bar{A}B + B\cdot\overline{\bar{A}\cdot B}]$

$= A\oplus[B\cdot(A+\overline{B})] = A\oplus[AB]$

A logical circuit is as given below:

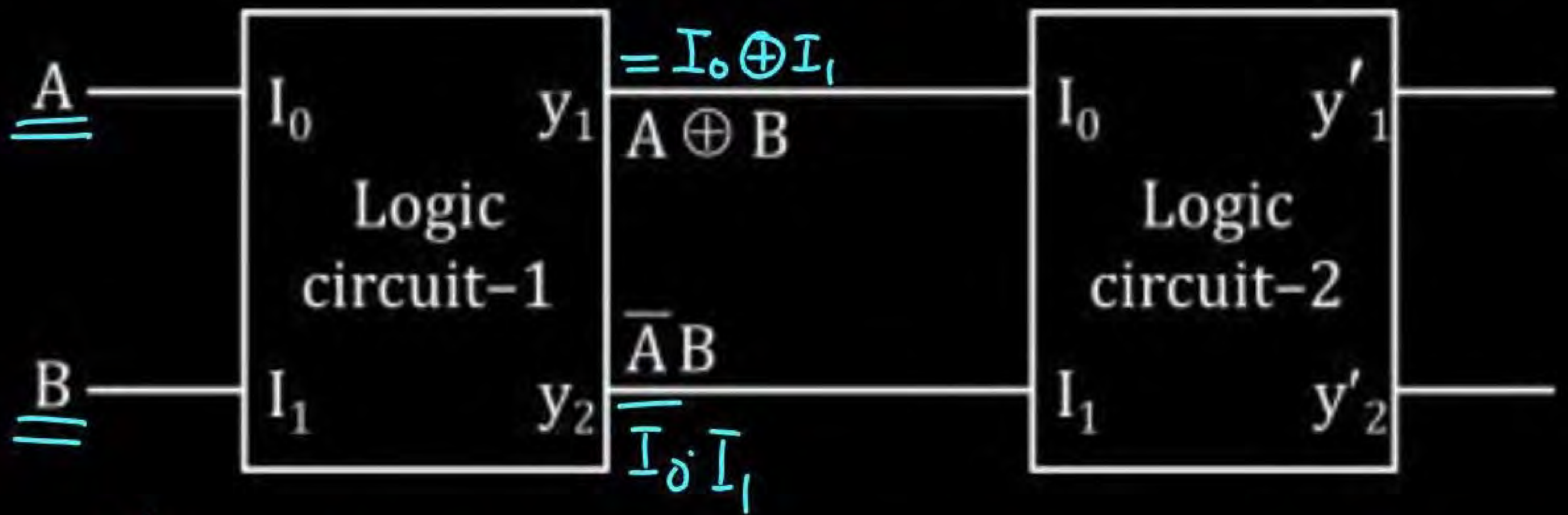
Function of logic circuit – 1 & logic circuit – 2 is same. Then output y'_1 will be

A $A \oplus B$

B $A\overline{B}$

C $\overline{A}B$

D $A - B$



$y_1' = A\oplus AB = \bar{A}\,AB + A\cdot\overline{A\cdot B} = A\cdot[\bar{A}+\bar{B}]$

$= 0 + A\bar{B} = A\bar{B}$

Question (MCQ) EASY

We have two numbers A and B and both are 4-bit numbers. To perform complete addition of these two numbers, total number of H.A.s + OR gate require is 10.

$A = a_3 a_2 a_1 a_0$,

$B = b_3 b_2 b_1 b_0$

$C_2 \; C_2 \; C_1$

$C_3 \mid S_3 \mid S_2 \mid S_1 \mid S_0$

S_0: 1 H.A.

S_1: 2 H.A + 1 OR

S_2: 2 H.A + 1 OR

S_3: 2 H.A + 1 OR

N-bit

(2N-1) H.A

(N-1) OR

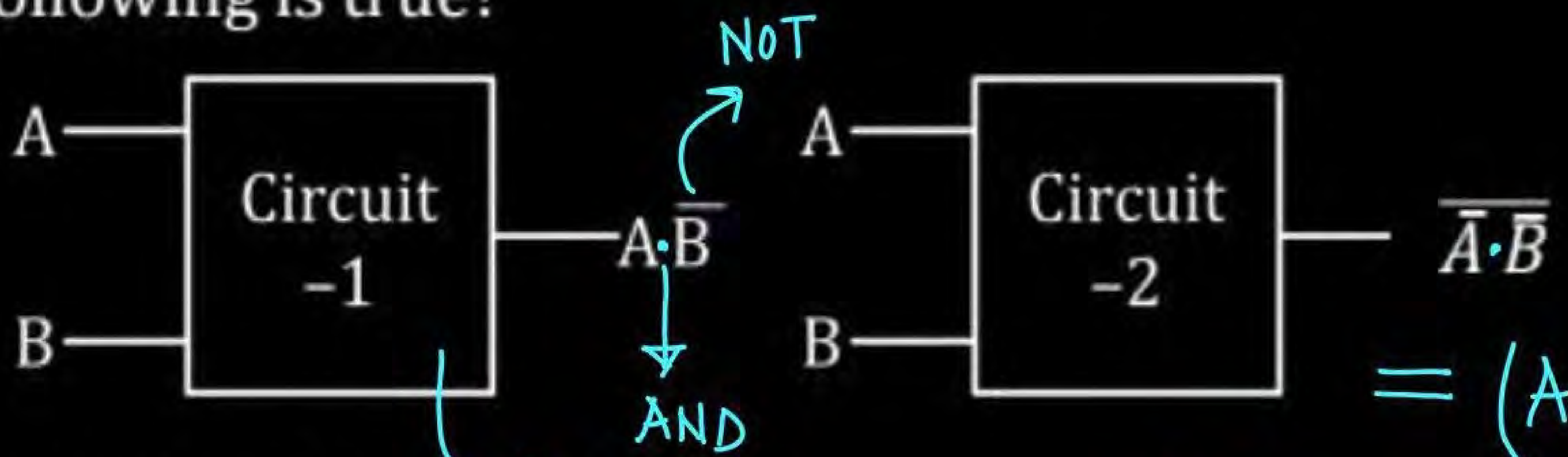
H.As $n = 7$ H.A

OR gate $m = 3$ OR

$n + m = 7 + 3 = 10$

Question (MCQ) BRAIN TEASER

Which of the following circuits can be used as universal circuit :
Then which of the following is true?



$= (A+B)$

A circuit – 1 and circuit – 2 both → universal

B circuit – 1 only

C circuit – 2 only

D Neither circuit – 1 nor circuit – 2

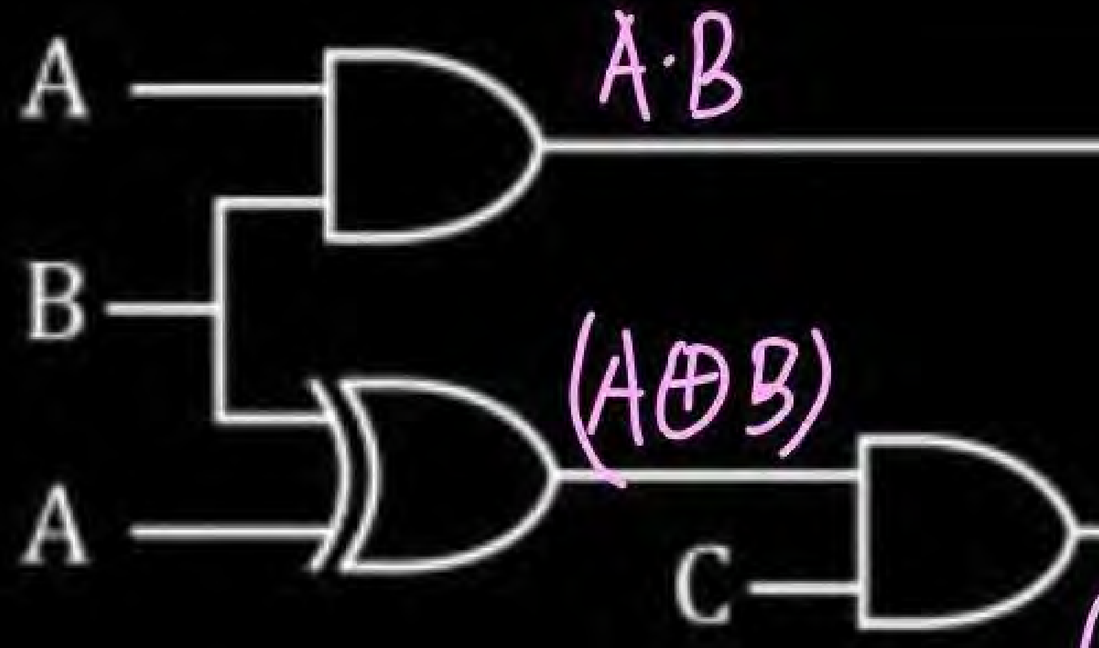
$\overline{\bar{A}\cdot\bar{0}} = \bar{\bar{A}} = A$

$\overline{\bar{A}\cdot\bar{1}} = \bar{0} = 1$

$\overline{\bar{A}\cdot\bar{A}} = \bar{\bar{A}} = A$

Question (MSQ) BRAIN TEASER

A logical circuit is implemented as : (A, B)
Output y is



$y = AB + (A \oplus B)C$

Carry O/P = $AB + BC + CA$

A) '1' when majority of the i/P$_s$ A, B, C are at '1'

B) '0' when minority of the i/P$_s$ A, B, C are at '1'

C) '1' when minority of the i/P$_s$ A, B, C are at '1'

D) '0' when majority of the i/P$_s$ A, B, C are at '1'

1,

A	B	C
1	0	0
0	1	0
0	0	1
0	0	0

Question (MCQ) BRAIN TEASER

x, y, z are i/Ps of a full adder then, sum output of full adder will be '1' if

$$\text{Sum} = x \oplus y \oplus z = \Sigma(1,2,4,7)$$

A $\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x} = 1$ ✗

B $\boxed{xyz = 1}$ ✗

C $(x + y)(y + z)(z + x) = 0$ ✗

D None of these ✓

Sum = 1 ← { only one '1' & two zeros; all 3 i/P are at 1.

$$\overline{(x+y)\cdot(y+z)\cdot(z+x)} = \bar{0}$$

$$\overline{x+y} + \overline{y+z} + \overline{z+x} = 1$$

$$(\bar{x}\cdot\bar{y}) + (\bar{y}\cdot\bar{z}) + \bar{z}\cdot\bar{x} = 1$$

Question (MSQ) (B, C) EASY

$\overline{\overline{A}\cdot\overline{B}}$

$= A + B$

Which of the following is/are true?

A NOR gate is equivalent to NAND gate with bubbled i/P_s.

B OR gate is equivalent to NAND gate with bubbled i/P_s.

C AND gate is equivalent to NOR gate with bubbled i/P_s.

D NAND gate is equivalent to NOR gate with bubbled i/Ps.

$\overline{\overline{A}+\overline{B}}$

$= A \cdot B$

Question (MCQ) MODERATE

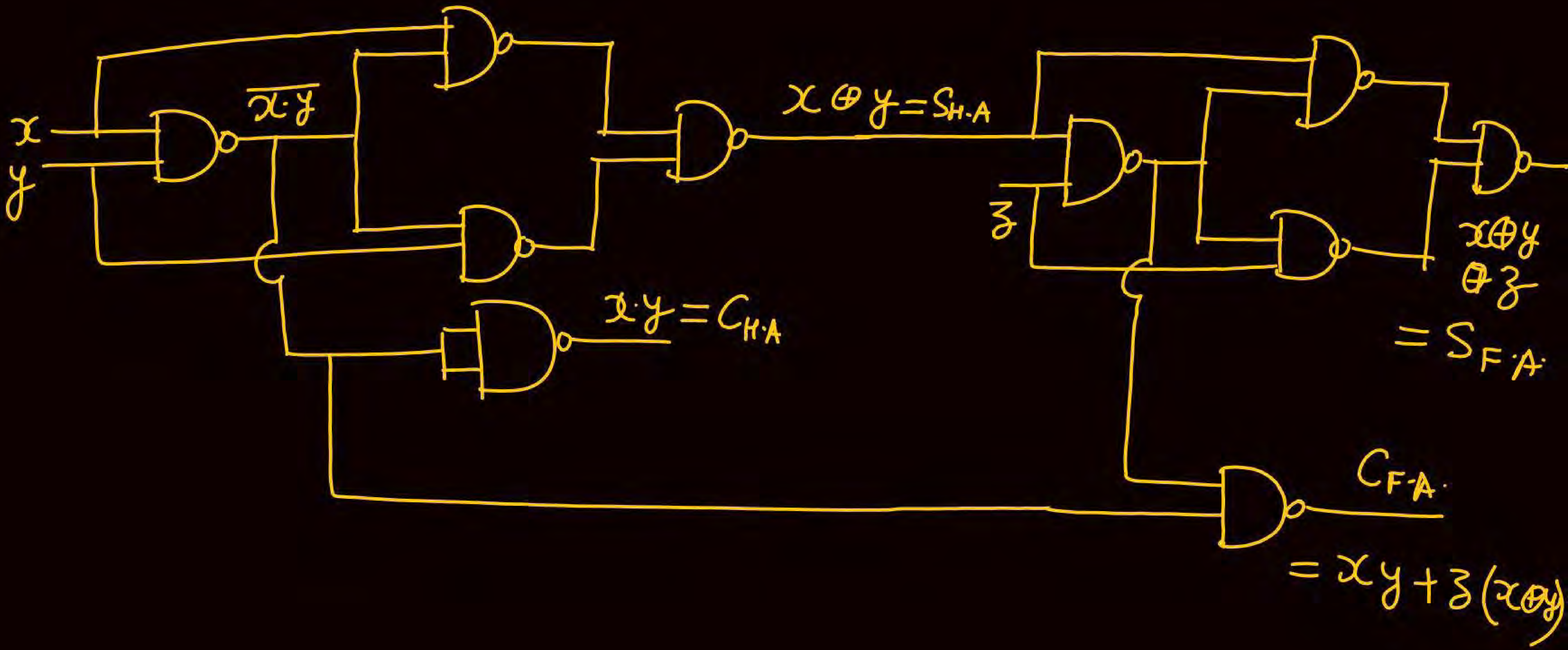
We have to implement H.A circuit & full adder circuit together. i/P lines for H.A. is x, y and i/P lines for full adder is x, y, z then minimum number of 2 – i/P NAND gate require is ___10___.

$$H.A \rightarrow S_{H.A} = x \oplus y$$

$$C_{H.A} = xy$$

$$F.A \rightarrow S_{F.A} = x \oplus y \oplus z$$

$$C_{F.A} = xy + (x \oplus y) \cdot z = xy + yz + zx.$$



x
y
$\overline{x \cdot y}$
$x \oplus y = S_{H \cdot A}$
$x \cdot y = C_{H \cdot A}$
z
$x \oplus y \oplus z = S_{F \cdot A}$
$C_{F \cdot A}$
$= xy + z(x \oplus y)$

Question NAT

We have a 4-bit adder in which each full adder has sum delay of t_s = 4 nsec and carry delay of 2 nsec. In $\boxed{\text{one minute}}$ number of additions completed by this full adder is 6×10^9.

$t_s = 4\,nsec,\ t_c = 2\,nsec$

Worst case delay $t_W = (n-1)t_c + \max[t_c, t_s] = 3t_c + \max(t_c, t_s)$

$= 3 \times 2 + \max(2ns, 4ns] = 10\,nsec$

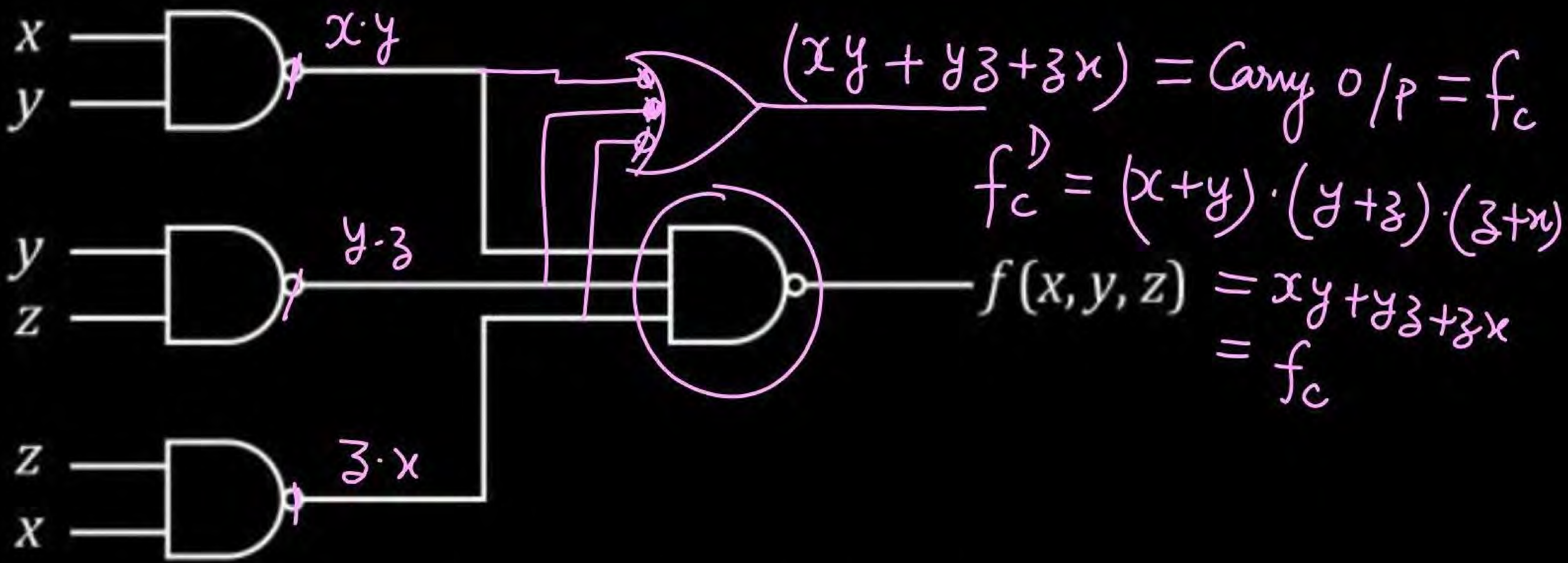
$\therefore$ 1 addition is performed in 10 nsec

$\therefore$ no. of addition performed in one $= \dfrac{1\,sec}{10\,nsec} = \dfrac{10^9}{10} = 10^8$ additions

$\therefore$ no. of addition in 1 minute (60 sec) $= 60 \times 10^8 = 6 \times 10^9$ additions

Question MSQ (A, C)

A logical circuit is as given below:



Which of the following circuit is/are equivalent to the above circuit:

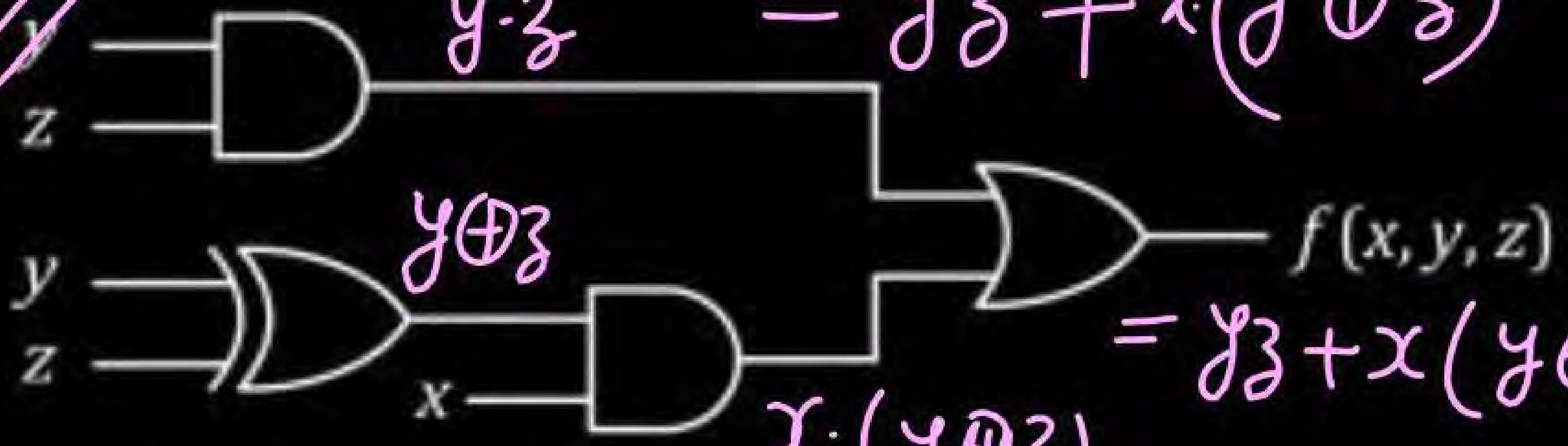
i/p

$\underline{y, z,}\ x$ previous bit generated carry $= yz + zx + xy$

A

$y \cdot z$ $\quad = yz + x \cdot (y \oplus z)$

$y \oplus z$



$= yz + x(y \oplus z)$

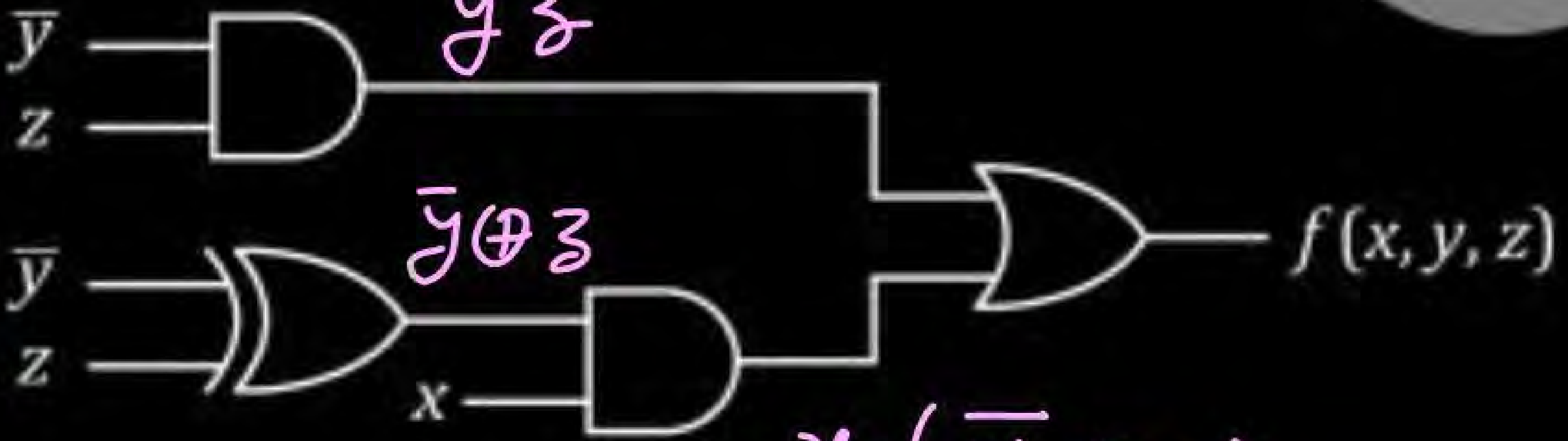
$x \cdot (y \oplus z)$

B

X

$\bar{y} z$

$\bar{y} \oplus z$

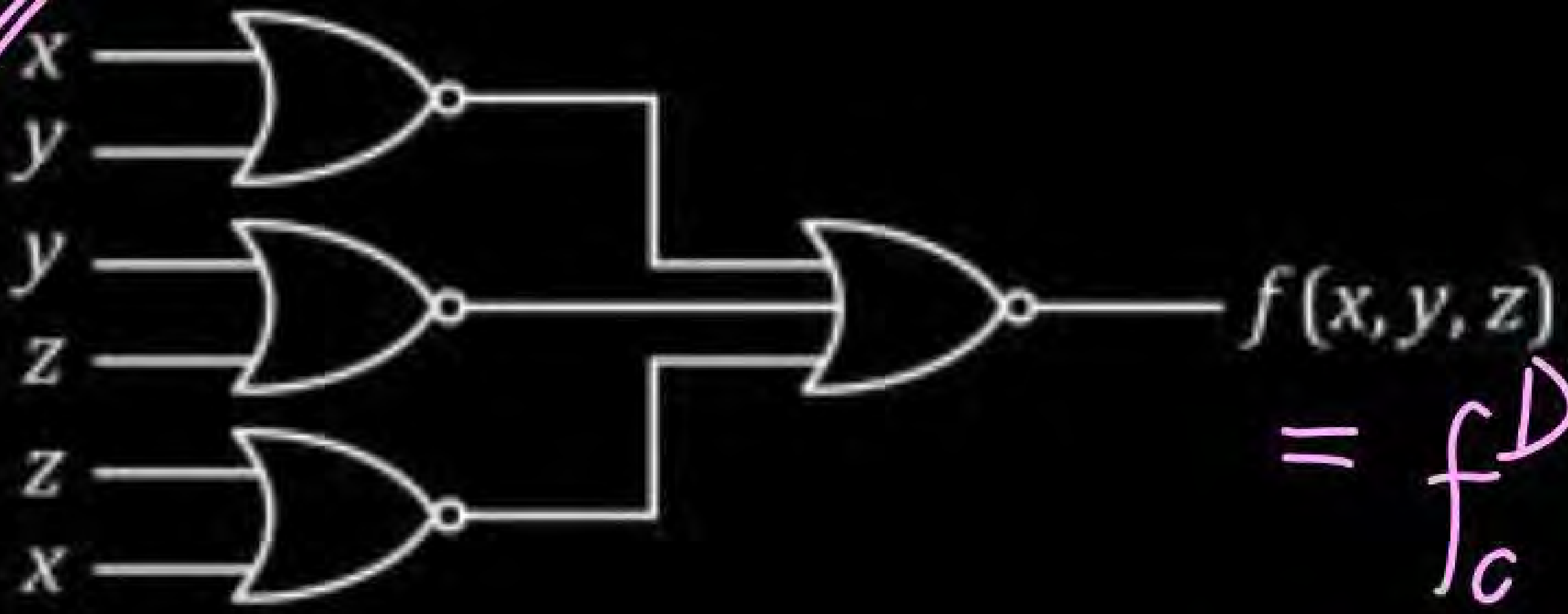


$x \cdot (\bar{y} \oplus z)$

$= \bar{y} z + x \cdot (y \odot z)$

$(y - z)$ i/p borrow bit x

C

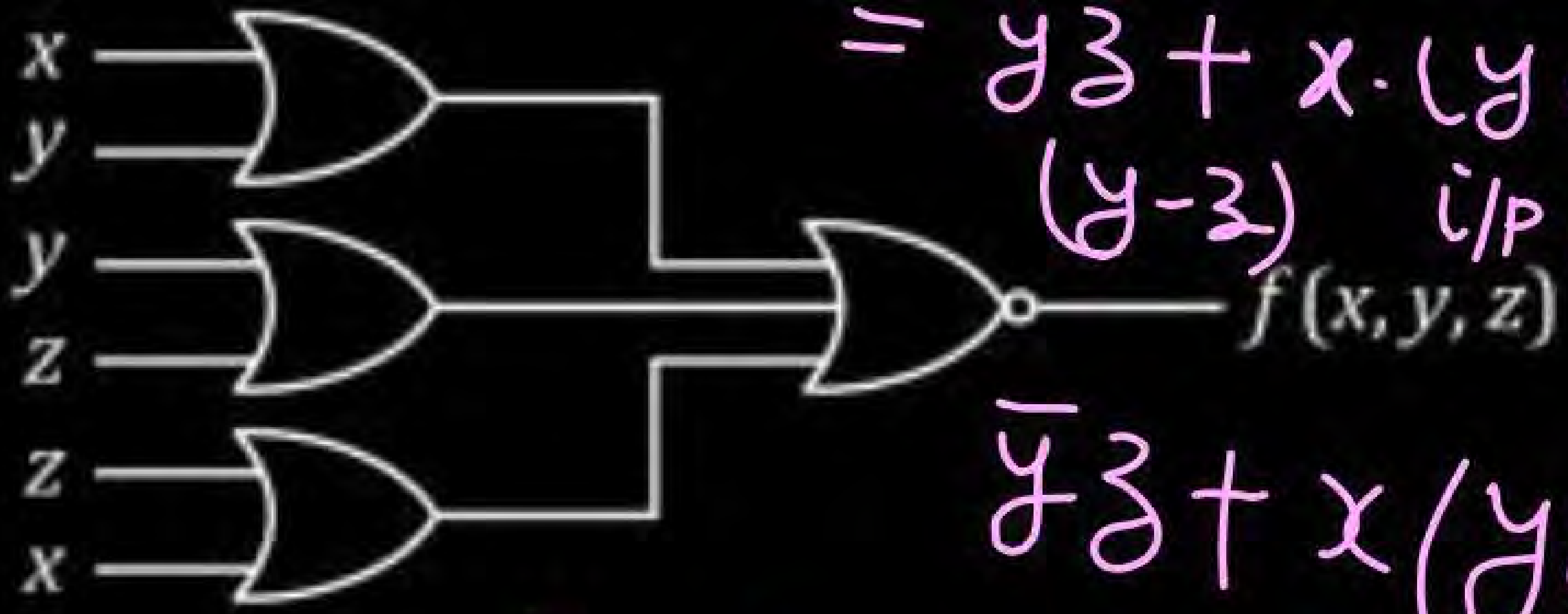


$= f_c^D$

$= f_c$

D

X



$\bar{y} z + x(y \odot z)$

$y \quad z \quad x$

Question MCQ

A K-map is as given below :

$\bar{x}\bar{y}$

$\bar{x}\bar{z}$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		1
x		1	1	

xz

In minimized solution of above K-map, there are terms $\bar{x}\bar{y}$, $\bar{x}\bar{z}$ and one term is missing. The missing term is

A $\bar{y}z$

B xz

C $\bar{x}z$

D $\bar{y}\bar{z}$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		1
x		1	1	

$\bar{x}\bar{z}, xz, \bar{y}z$

or $\bar{x}\bar{z}, xz, \bar{x}\bar{y}$

Question

MCQ

A K-Map is as given below:

$\bar{x}\bar{y}$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		X
x	X		1	1

$\bar{z}$

xy

$= \bar{x}\bar{y} + xy$

$= x \odot y$

It minimized solution will be

A $x \odot y$

B $x \oplus y$

C $\bar{z} + \bar{x}\bar{y}$

D $\bar{z} + xy + \bar{x}\bar{y}$

Question

MCQ

$F(x,y,z) = \overline{x\bar{y} + \bar{x}y} + x\bar{y}z + x\bar{y}\bar{z}$ $= \overline{f_1} + x\bar{y}z + x\bar{y}\bar{z}$

Then the minimized solution of F(x,y,z) will be

$f_1 = x\bar{y} + \bar{x}y$

$F = x + \bar{y}$

$\overline{f_1}$	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1	0	0
x	0	0	1	1

F	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		
x	1	1	1	1

$\bar{y}$ x

A $x + \bar{y}$

B $\bar{x}y$

C $xy\bar{z}$

D $x + \bar{y} + z$

Question

NAT

★★☆

In a 4-bit parallel adder, sum delay and carry delay of each full adder is $t_s = 1$ nsec and $t_c = 5$ nsec respectively. Number of additions performed per sec by adder is N_1. If delays are interchanged then no. of additions performed per sec is N_2 then N_2/N_1 ___2.5___.

$n = 4$ bits $\longrightarrow t_s = 1\,ns,\ t_c = 5\,nsec \longrightarrow N_1$ additions in one second

$\therefore \frac{N_2}{N_1} = \frac{10^9/8}{10^9/20} = 2.5$

Worst delay $t_W = (n-1)t_c + \max(t_c, t_s) = 3t_c + \max(5ns, 1ns) = 20\,nsec$

$N_1 = \frac{1\,sec}{t_W} = \frac{1\,sec}{20\,nsec} = \frac{1}{20 \times 10^{-9}} = \frac{10^9}{20}$ additions

$t_s' = 5\,nsec,\ t_c' = 1\,nsec \rightarrow t_W' = 3t_c' + \max(t_c', t_s') = 3 \times 1 + \max(1\,nsec, 5\,ns) = 8\,nsec$

$\therefore N_2 = \frac{1\,sec}{t_W'} = \frac{1\,sec}{8\,nsec} = \frac{10^9}{8}$ additions

Question

MCQ

A K-map is as given below :

$\bar{A}\bar{C}$ $\quad$ $\bar{A}D$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	
$\bar{A}B$	1	1	1	X
AB			1	
$A\bar{B}$		X	X	

CD

$= \bar{A}\bar{C} + CD$

Then minimized solution of K-map is :

A $\bar{A}\bar{C}+\bar{A}D+CD$ ✗

B $\bar{A}\bar{C}+CD$

C $\bar{A}B+CD+\bar{A}\bar{C}$

D None of these

Question

MCQ

K-map of two logical functions f_1 (A, B) and f_2 (A, B) is as given below :

f_1	$\overline{B}$	B
$\overline{A}$	1 $_0$	$_1$
A	$_2$	1 $_3$

$f_1(A,B) = \Sigma(0,3)$

f_2	$\overline{B}$	B
$\overline{A}$	1 $_0$	1 $_1$
A	$_2$	$_3$

$f_2(A,B) = \Sigma(0,1)$

Then K-map of f (A, B) will be

$A=0, B=0$

$A=0, B=1$

$= \Sigma(1,2,3)$

A

f	$\overline{B}$	B
$\overline{A}$	1	1
A		1

B

f	$\overline{B}$	B
$\overline{A}$		1
A	1	1

C

f	$\overline{B}$	B
$\overline{A}$	1	1
A		

D

f	$\overline{B}$	B
$\overline{A}$		1
A	1	

Question

NAT

f (A, B, C, D) is as given below :

$$f(A,B,C,D)=\overline{\bar{B}D+AC+\bar{A}B}+ABCD+\bar{A}BC+ABD = \overline{f_1}+ABCD+\bar{A}BC+ABD$$

Then number of entries in K-map of $\bar{f}$ (A, B, C, D) that are filled with '1' ___8___.

$f_1=\bar{B}D+AC+\bar{A}B$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1			1
$\bar{A}B$			1	1
AB	1	1	1	
$A\bar{B}$	1			

$\bar{f}_1$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	0	0	1
$\bar{A}B$	0	0	0	0
AB	1	1	0	0
$A\bar{B}$	1	0	0	0

Question

MCQ

f (x, y, z) is given as : $f(x,y,z)=\bar{x}\bar{y}+\bar{x}z+xy$,
The equivalent f (x, y, z) can also be

A $(\bar{x}+y+\bar{z})(x+\bar{y}+\bar{z})(x+\bar{y}+z)$

B $(x+\bar{y}+z)(\bar{x}+y+z)(\bar{x}+y+\bar{z})$ ✓

C $(\bar{x}+\bar{y})(\bar{x}+z)(x+y)$ ✗

D None of these

f

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1 — 0	1 — 1	1 — 3	0 — 2
x	0 — 4	0 — 5	1 — 7	1 — 6

$001 \rightarrow \bar{x}\bar{y}z$

$$f(x,y,z)=\Sigma(0,1,3,6,7)=\pi(2,4,5)$$

010, 100, 101

$$=(x+\bar{y}+z)\cdot(\bar{x}+y+z)(\bar{x}+y+\bar{z})$$

Question

MCQ

Which of the following is true ?

A. K-map has more than one solution. X

B. K-map has always unique solution. X

C. K-map can have more than one solution.

D. None of these

Question NAT

uestion

$A > B = 1+2+3 = 6$

Two no's A & B are two bit no's as

$A = a_1a_0$ and $B = b_1b_0$

Then no. of combination in which $A > 2B$ 4.

$A \rightarrow 2$ bit $\rightarrow 0, 1, 2, 3$

$B \rightarrow 2$ bit $\rightarrow 0, 1, 2, 3$

$2B \rightarrow 0, 2, 4, 6$

When $A = 0 \rightarrow$ no combination $A > 2B$

When $A = 1 \rightarrow$ one combination $A > 2B$ [when $2B = 0$]

When $A = 2 \rightarrow$ one combination $A > 2B$ [$2B = 0$]

When $A = 3 \rightarrow$ two combinations $A > 2B$ [$2B = 0$, $2B = 2$]

Total $= 1+1+2 = 4$

Question NAT ★★★

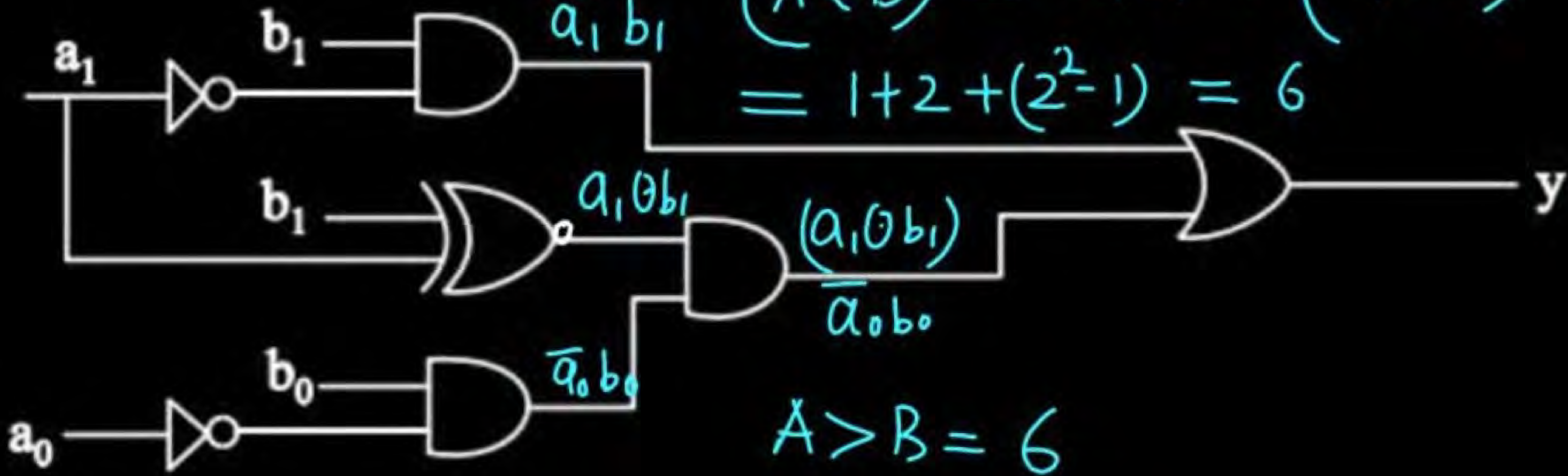
A digital ckt is as given below:

$A = a_1 a_0$

$B = b_1 b_0$

$(A<B) = \bar{a}_1 b_1 + (a_1 \odot b_1)\, \bar{a}_0 b_0$

$= 1 + 2 + (2^2 - 1) = 6$



$y = \bar{a}_1 b_1 + (a_1 \odot b_1)\, \bar{a}_0 b_0$

$A>B = 6$

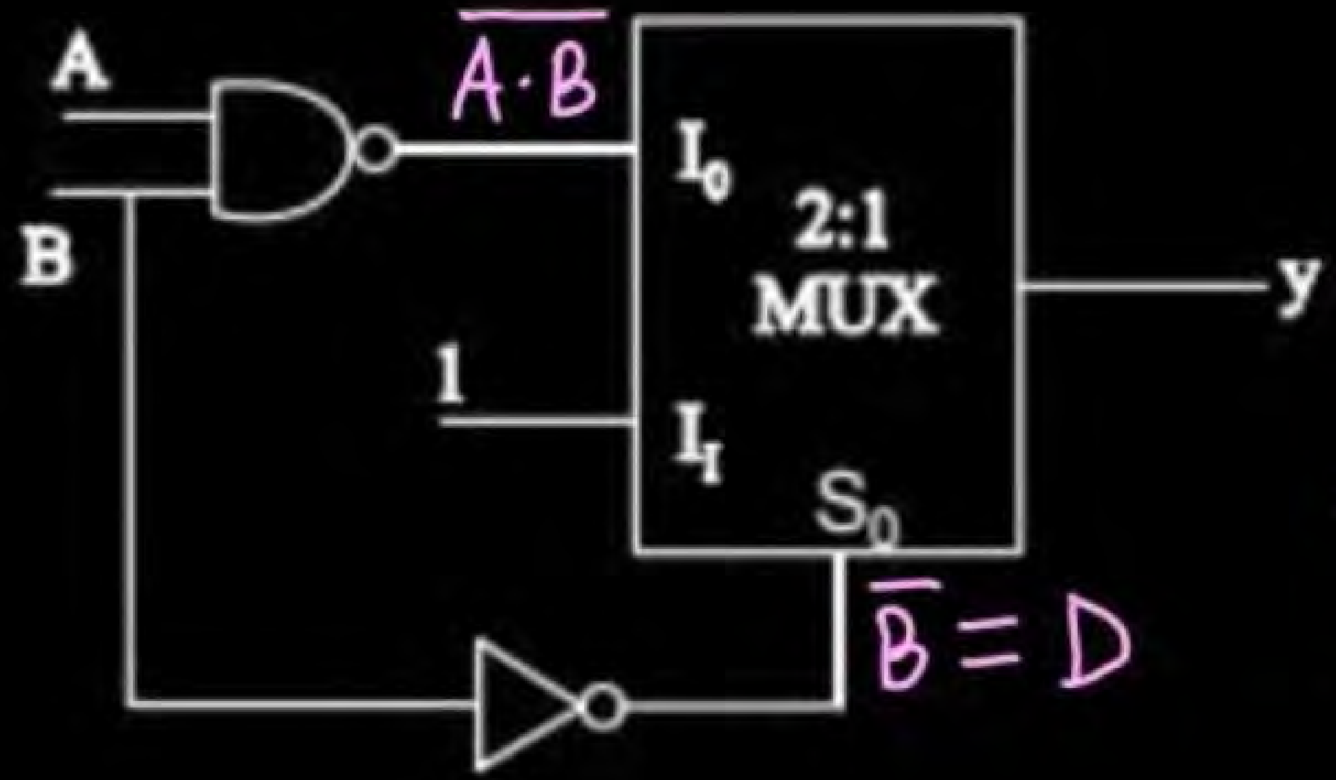
$A=B = 2^n = 4$

i/p lines of the ckt are a_1, b_1, a_0, b_0 and o/p line is y.
Then no. of combinations of i/p lines in which o/p y will be '1' is 6.

Question MCQ

estion

A MUX circuit is as given below:



$y = \bar{D}\cdot\overline{A\cdot B} + D\cdot 1$

$= B(\bar{A}+\bar{B}) + \bar{B}$

$= \bar{A}\cdot B + 0 + \bar{B}$

$= \bar{A}B + \bar{B} = \bar{B} + (\bar{A}\cdot B)$

$= (\bar{B}+\bar{A})\cdot(\bar{B}+B) = (\bar{A}+\bar{B})$

$= \overline{A\cdot B}$

o/p y is.

A $\bar{A}+\bar{B}$

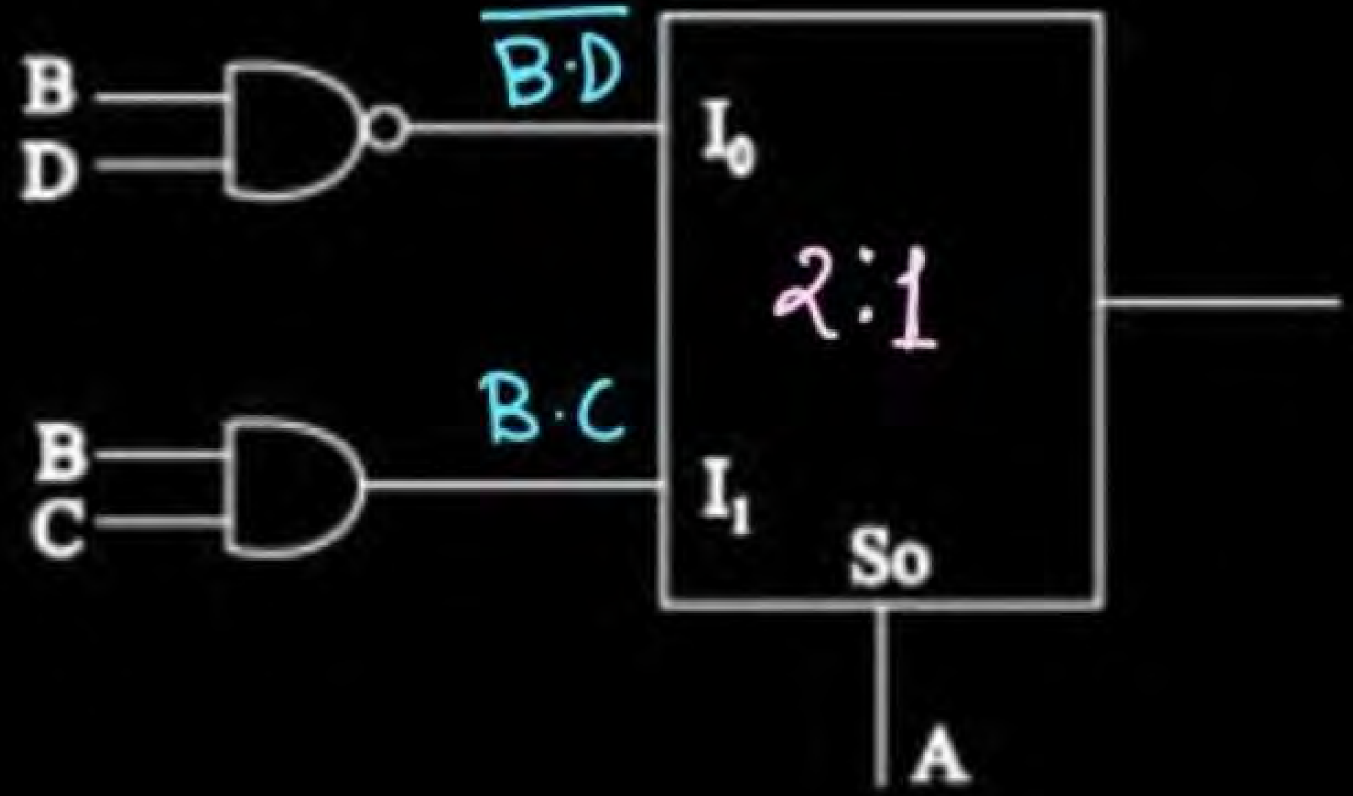
B $\overline{A+B}$

C $A \oplus B$

D $A\bar{B}$

Question **MCQ**

A MUX ckt is as given below:



$$f(A, B, C, D) = \bar{A}\,\overline{B\cdot D} + A\cdot BC$$
$$= \bar{A}(\bar{B}+\bar{D}) + ABC$$
$$= \bar{A}\bar{B} + \bar{A}\bar{D} + ABC$$

f(A, B, C, D) is:

A $\Sigma(0, 1, 2, 3, 4, 5, 6, 14, 15)$

B $\Sigma(0, 1, 2, 3, 4, 6, 13, 15)$

C $\Sigma(0, 1, 2, 3, 4, 6, 14, 15)$

D $\Sigma(1, 2, 3, 4, 5, 6, 7, 13, 14, 15)$

$\bar{A}\bar{B}$

$0\ 0\ 0\ 0 \longrightarrow 0$

$0\ 0\ 0\ 1 \longrightarrow 1$

$0\ 0\ 1\ 0 \longrightarrow 2$

$0\ 0\ 1\ 1 \longrightarrow 3$

$\bar{A}\quad\quad \bar{D}$

$0\ 0\ 0\ 0 \longrightarrow 0$

$0\ 0\ 1\ 0 \longrightarrow 2$

$0\ 1\ 0\ 0 \longrightarrow 4$

$0\ 1\ 1\ 0 \longrightarrow 6$

$\Sigma 0, 1, 2, 3, 4, 6, 14, 15)$

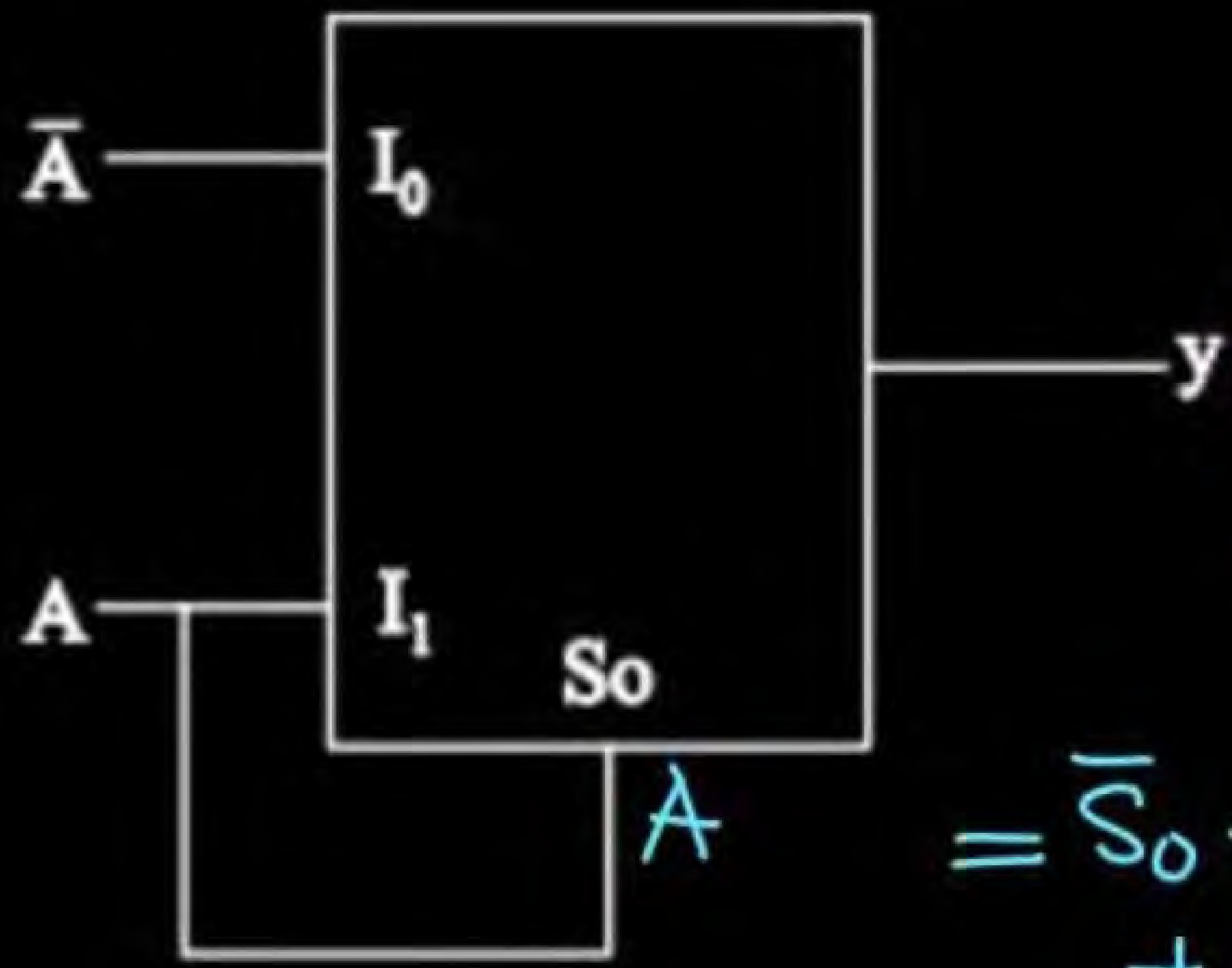
$A\ B\ C$

$1\ 1\ 1\ 0 \longrightarrow 14$

$1\ 1\ 1\ 1 \longrightarrow 15$

Question MCQ

A MUX ckt is as given below
o/p y is



$y = A \odot S_0$
$= A \odot A$
$= 1$

$= \bar{S_0} \cdot \bar{A} + S_0 \cdot A$
$= \bar{A} \cdot \bar{A} + AA$
$= A + \bar{A} = 1$

A $\bar{A}$

B A

C 0

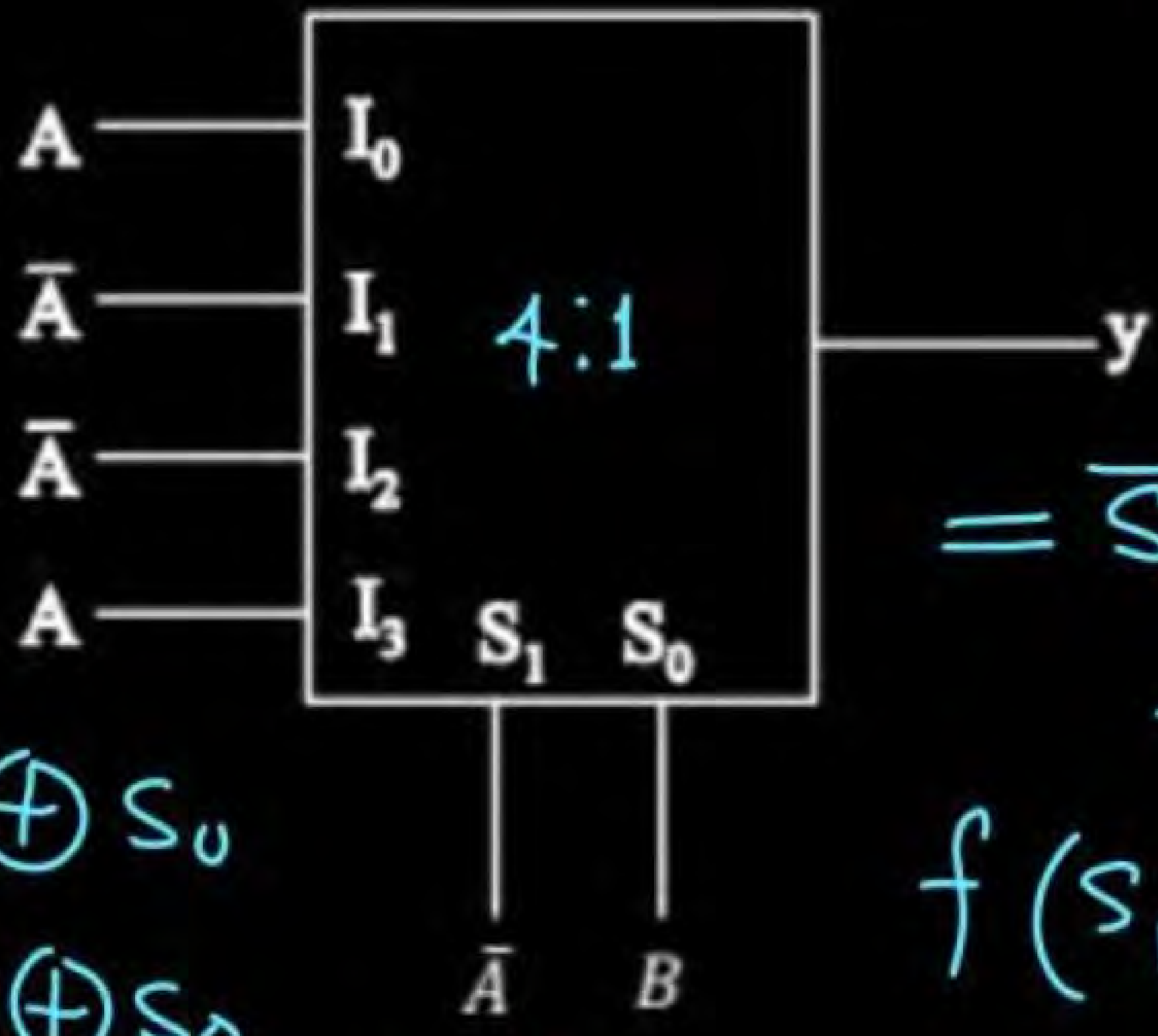
D 1

Question **MCQ**

$y = 1 \oplus B = \bar{B}$

A 4:1 MUX is given as:

D	$\bar{x}$	C
$\bar{D}$	x	$\bar{C}$
$\bar{D}$	x	$\bar{C}$
D	$\bar{x}$	C



$y = C \oplus S_1 \oplus S_0 = A \oplus \bar{A} \oplus B$

$= \bar{S}_1 \bar{S}_0 \cdot C + \bar{S}_1 S_0 \cdot \bar{C} + S_1 \bar{S}_0 \bar{C} + S_1 S_0 \cdot C$

$f(S_1, S_0, C) = \Sigma(1, 2, 4, 7)$

$= S_1 \oplus S_0 \oplus C$

$\bar{x} = D$

$x = \bar{D}$

$= D \oplus S_1 \oplus S_0$

$= \bar{x} \oplus S_1 \oplus S_0$

$= \overline{x \oplus S_1 \oplus S_0}$

o/p y is

A $\bar{B}$

B B

C $A \oplus B$

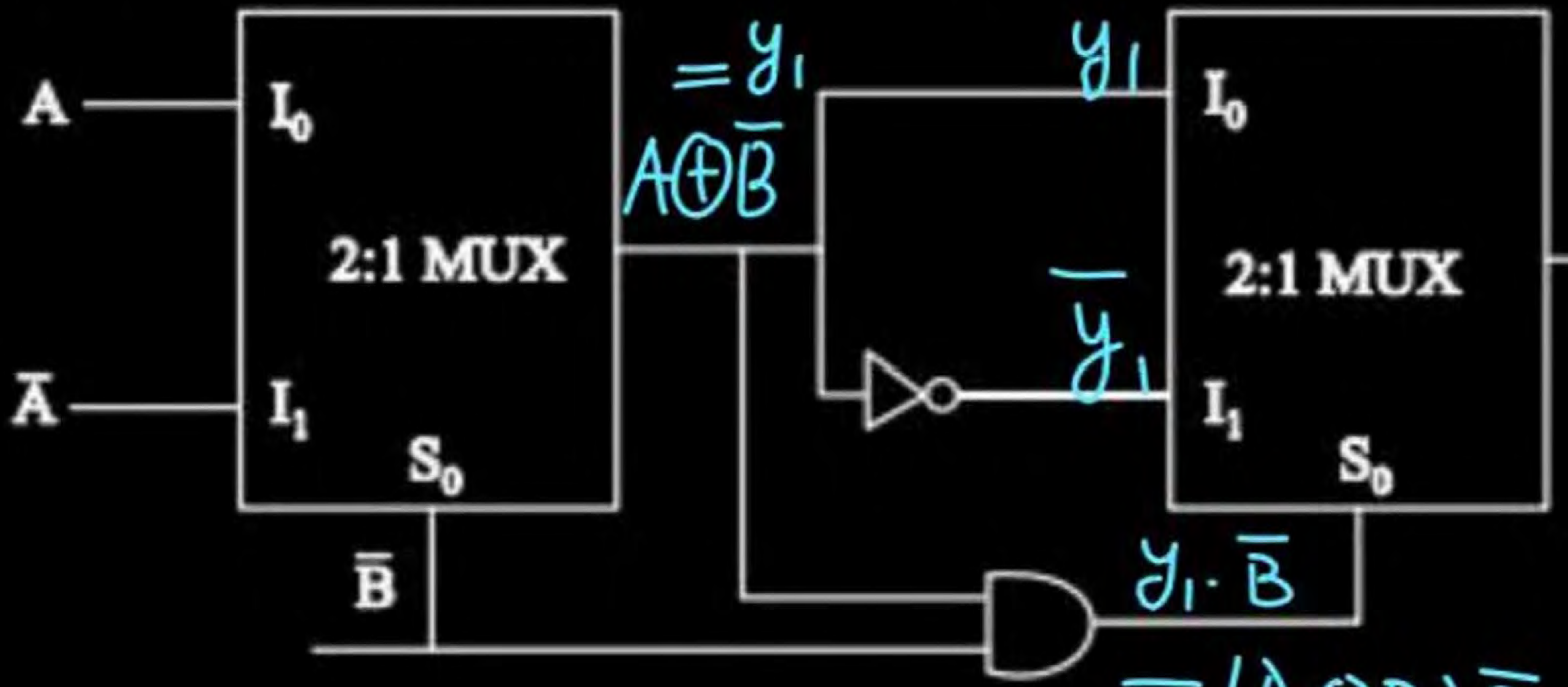
D $A \odot B$

Question MCQ ★★★

$A \oplus B \oplus AB = (A+B$ [redacted]

$A \oplus B \oplus (A+B) = A \cdot B$

A combinational ckt is designed using MUX as shown below :-



$y = y_1 \oplus S_0 = A \oplus \bar{B} \oplus \overline{(A+B)}$

$= A \oplus B \odot \overline{(A+B)}$

$= A \oplus B \oplus (A+B)$

$= A \cdot B$

o/p y is

$= (A \odot B)\bar{B} = (AB + \bar{A}\bar{B})\bar{B} = \bar{A}\bar{B} = \overline{A+B}$

A $\overline{A+B}$

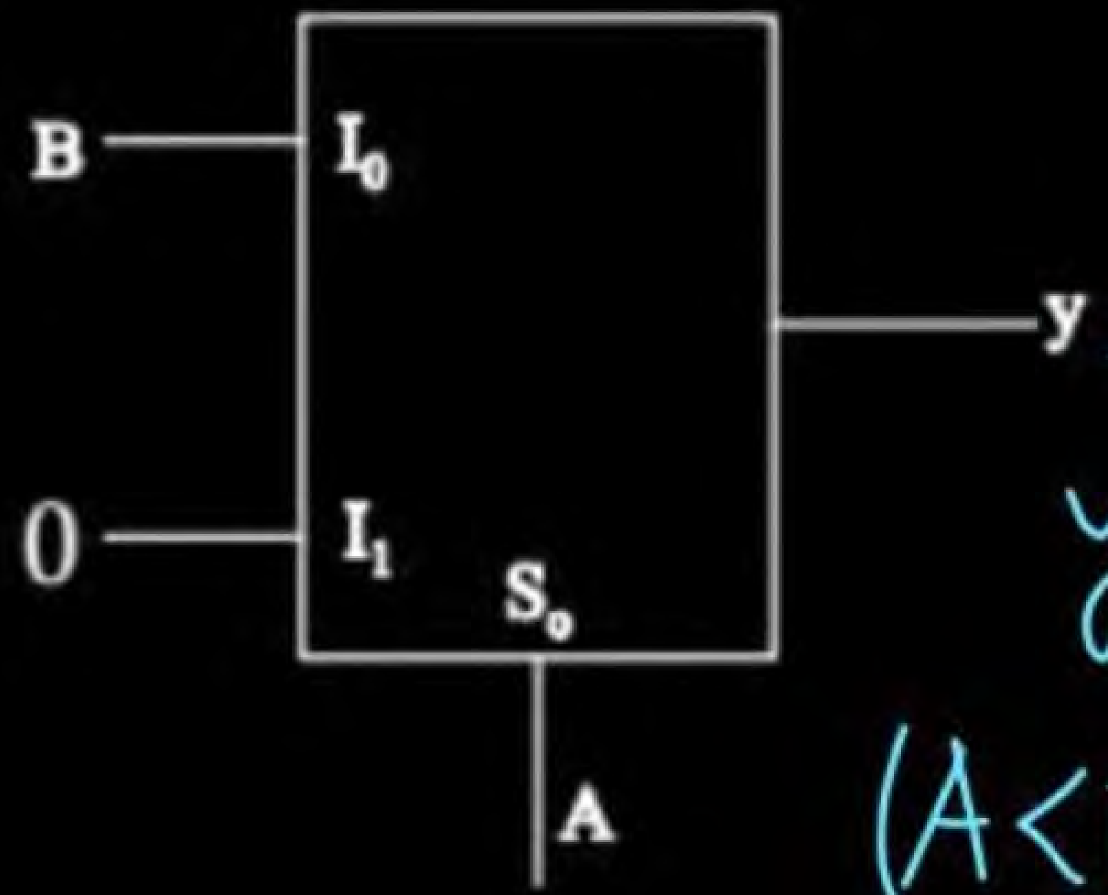
B A.B

C $A \oplus B$

D $A \odot B$

Question MCQ

A MUX circuit is implemented as shown:



$y = \bar{A}B + A\cdot 0$

$y = \bar{A}\cdot B$

$(A<B) = \bar{A}B$

$(A>B) = A\cdot\bar{B}$

$(A=B) = A\odot B$

o/p y represents

A Comparator o/p when B < A

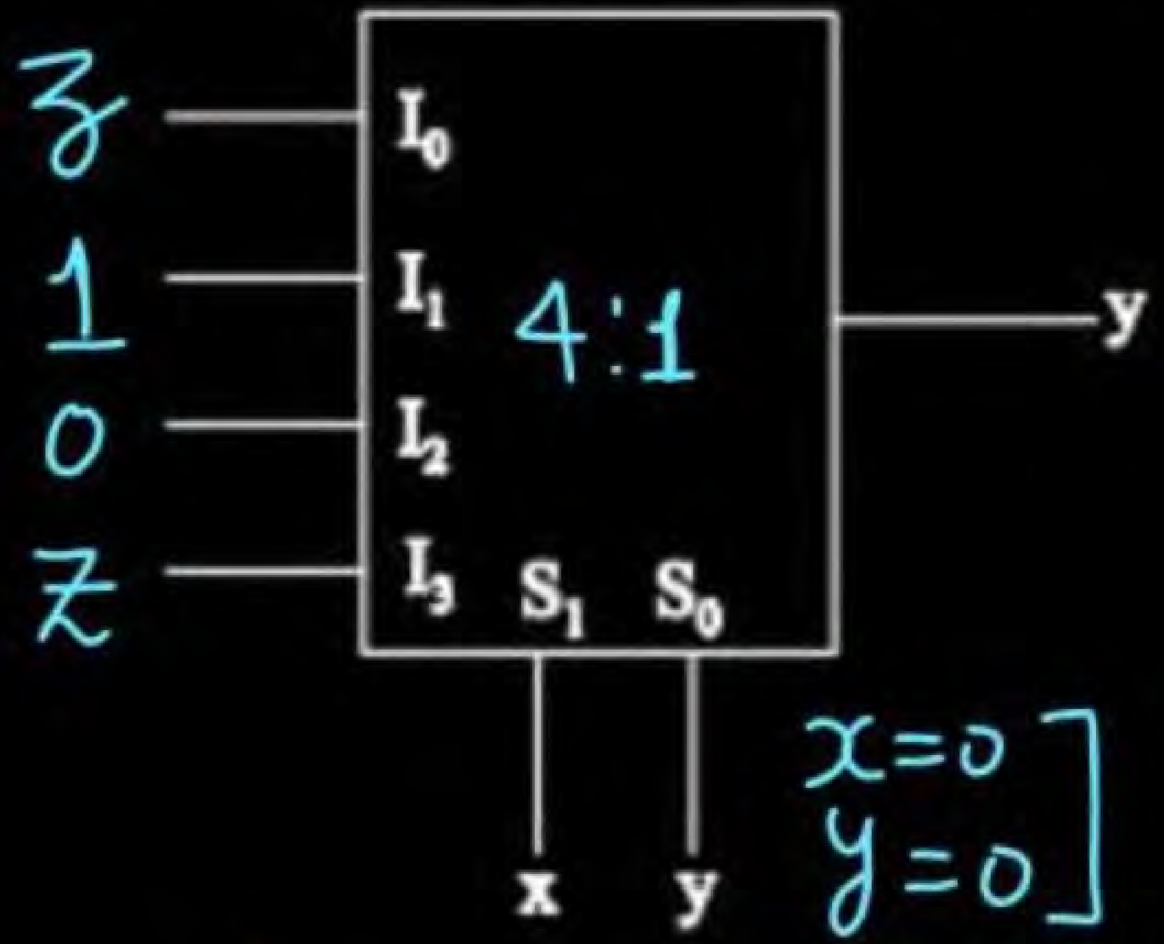
B Comparator o/p when B = A

C Comparator o/p when B > A

D Comparator o/p when B < 2A

Question MCQ

In a full subtractor corresponding i/p bits of two no.s are x & y [x – y] and input borrow is z and output borrow is B. Output borrow is implemented as shown :



$B = \bar{x}y + \bar{x}z + yz$

$B(x=0, y=0) = z + 0 = z$

$= \bar{x}y + \bar{x}z + yz$

$B(x=0, y=1)$

$= 1 + \text{—}$

$= 1$

$\left.\begin{matrix} x=0 \\ y=0 \end{matrix}\right]$ $\begin{matrix} x=0 \\ y=1 \end{matrix}$

Then values of I_0, I_1, I_2 & I_3 are :

A $I_0 = z, I_1 = 1, I_2 = 0, I_3 = z$

B $I_0 = z, I_1 = 0, I_2 = 1, I_3 = \bar{z}$

C $I_0 = 1, I_1 = z, I_2 = 0, I_3 = z$

D $I_0 = \bar{z}, I_1 = 1, I_2 = 0, I_3 = z$

Question NAT

★★☆

To implement 1024 : 1 MUX, how many 4 : 1 MUXs are required 341.

$$N = \frac{1024}{4} + \frac{256}{4} + \frac{64}{4} + \frac{16}{4} + 1$$

$$= 256 + 64 + 16 + 4 + 1$$

$$= 341$$

Question MCQ

Which of the following is true:

A MUX is non-universal ckt. X

B MUX can be used to implement half adder as well as full adder.

C X 2:1 MUX has 1–i/p line, 1–o/p line and 2-select lines.

D If in a MUX input lines are 24 then minimum 4-select lines are required for proper functioning.

$M \geq \log_2 N$ $\quad M \geq 4.g$

$M \geq \log_2 24$ $\quad M \geq 5$

Question MCQ

Which of the following is true ?

A Decoder and MUX has same internal circuitry

B MUX & De MUX has same internal circuitry

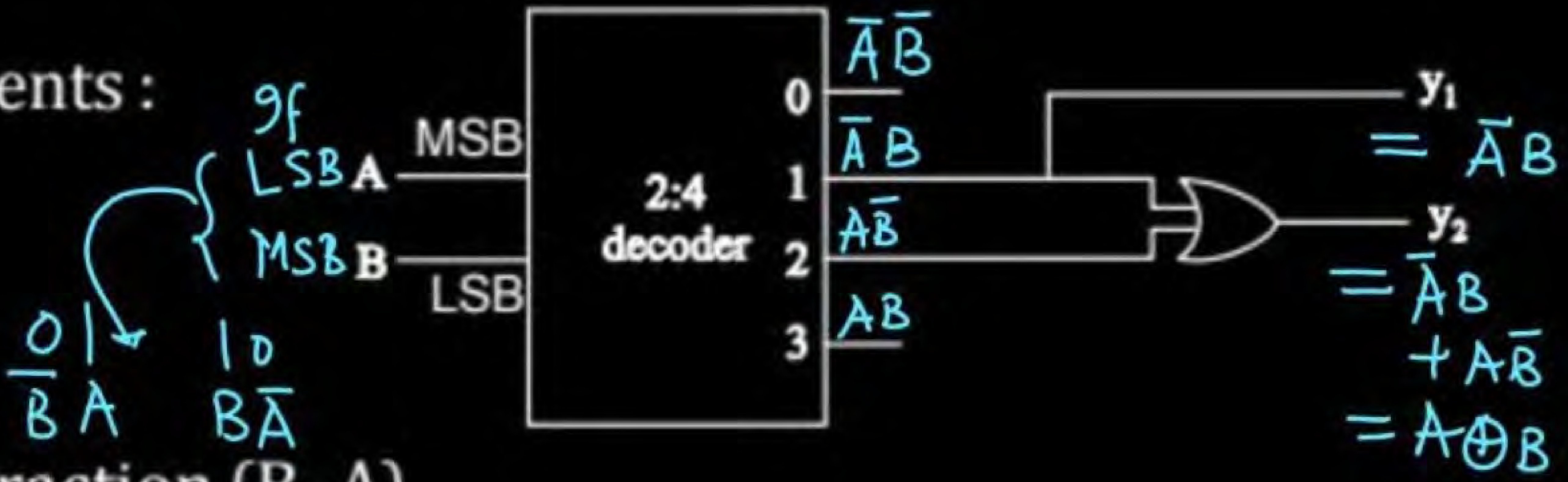
C De MUX and decoder has same internal circuitry

D None of these

Question MCQ

A combinational ckt using decoder and OR gate is as given below :

o/p y_1 & y_2 combinedly represents :



A Half adder

B Half subtractor with subtraction (B–A)

C Half subtractor with subtraction (A–B)

D Comparator o/p with (A > B)

Question NAT

★☆☆

To implement 1:512 De MUX, we require N, 1:8 De MUX, then the value of N is __73__.

$$N = \frac{512}{8} + \frac{64}{8} + \frac{8}{8}$$

$$N = 64 + 8 + 1$$

$$\boxed{N = 73}$$

Question MSQ (B, C)

* * *

$$1:128 = (1:2^7) \xrightarrow{1:2} (2^7-1) = 127$$

To implement 1:128 De MUX, we require

A 127 – 1:4 DeMUX ✗

B 127 – 1:2 DeMUX ✓

C 18 – 1:8 DeMUX & 1 – 1:2 DeMUX ✓

D 40 – 1:8 DeMUX & 1 – 1:2 DeMUX ✗

$$1:128 \xrightarrow{1:4} \underbrace{32 + 8 + 2}_{42(1:4)} + 1(1:2)$$

$$1:128 \xrightarrow{1:8} \underbrace{16 + 2}_{18(1:8) + 1(1:2)}$$

Question NAT

A combinational ckt is as given below

When i/p is A = 0 & B = 1, then LED no. that glows will be 2.

01 10

Question MCQ

* * *

For a decoder having 4 i/p lines, no. of o/p lines are

A 1

B 2

C 16

D 4

$n : 2^n$

Question MCQ

A digital ckt is as given below:

$$y = \bar{A}\cdot A\bar{B} + A\cdot AB$$
$$= 0 + AB = AB$$

o/p y is

A A

B B

C A.B

D (A + B)

Question **MCQ**

Which of the following is $\boxed{\text{not true'}}$?

$M \geq \log_2 N$

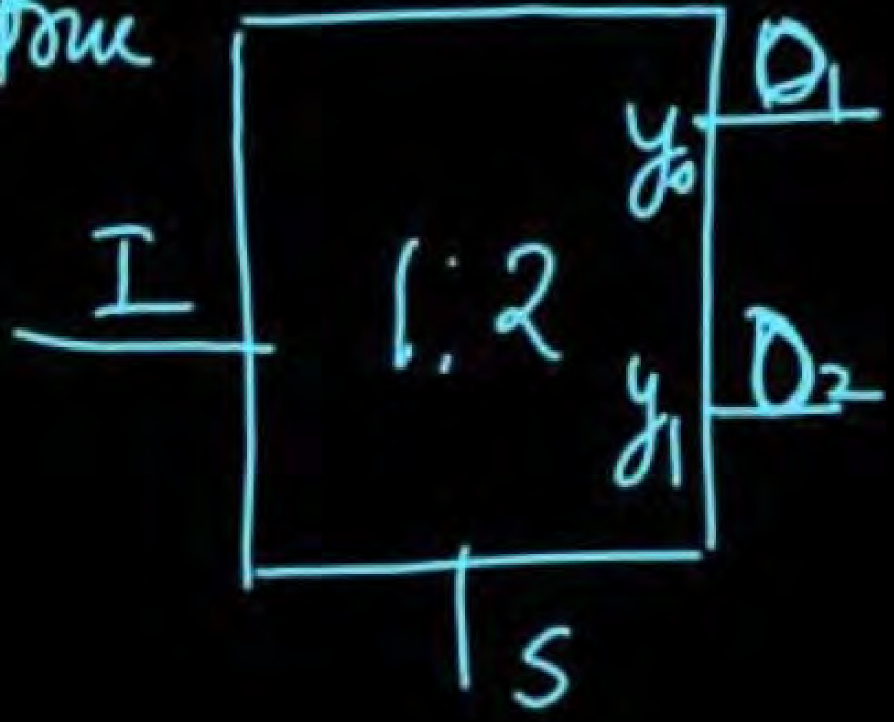
$M \geq \log_2 52 \rightarrow M \geq 5.7$

A DeMUX is not a universal ckt → true

B For 52 output lines, no. of select lines must be atleast 6 for DeMUX. — true

C DeMUX is parallel to serial converter. not true

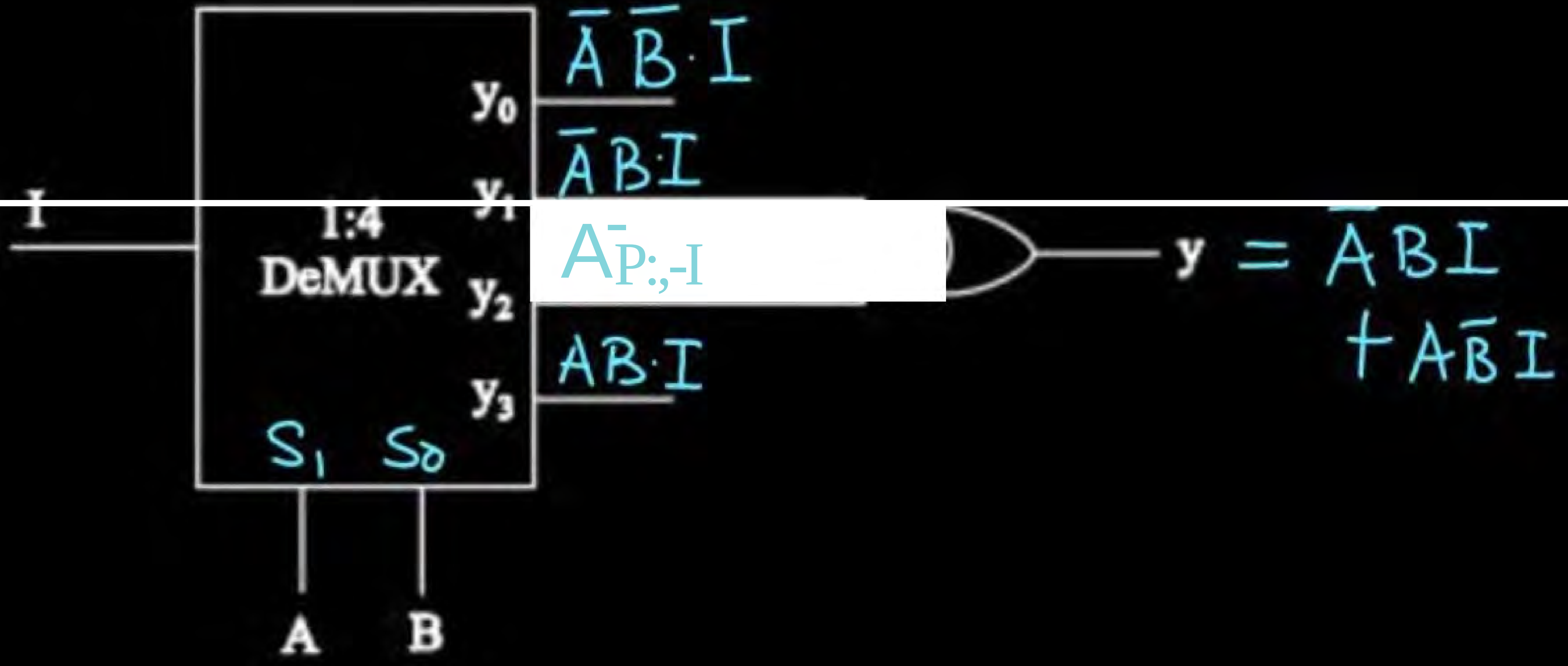
D DeMUX is serial to parallel converter true



Question MCQ

A ckt is as shown below:

o/p y is



A) $A \oplus B$ if $I = 1$

B) $A \odot B$ if $I = 0$ X

C) $A.B$ if $I = 1$ X

D) $A + B$ if $I = 0$ X

Question NAT

For implementing full adder and full subtractor we require a decoder of the order n:M, where n is no. of i/p lines & M is no. of o/p lines, then value of M is ___8___.

$$\boxed{n:M} \rightarrow 3:2^3$$

$$F.A \begin{cases} S = f(x,y,z) = \Sigma(1,2,4,7) \\ C = f(x,y,z) = \Sigma(3,5,6,7) \end{cases}$$

$$F.S \begin{cases} D = f(x,y,z) = \Sigma(1,2,4,7) \\ B = f(x,y,z) = \end{cases}$$

Question MSQ (C)

no. of i/p lines = 4

4-i/p lines XNOR →

It is given that:

$x_1 \odot x_2 \odot x_3 \odot x_4 = 0$ → at i/p side odd no. of 1's are there. → one 1's or three 1's

one i/p line is at '1' & other three are at '0'

Then which of the following is/are true?

$$\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix}$$

$\boxed{1\,0\,1\,0}$

A $x_1 x_2 + x_3 x_4 = 1$ ×

B × $x_1 \oplus x_2 + x_2 \oplus x_3 + x_3 \oplus x_4 = 1$

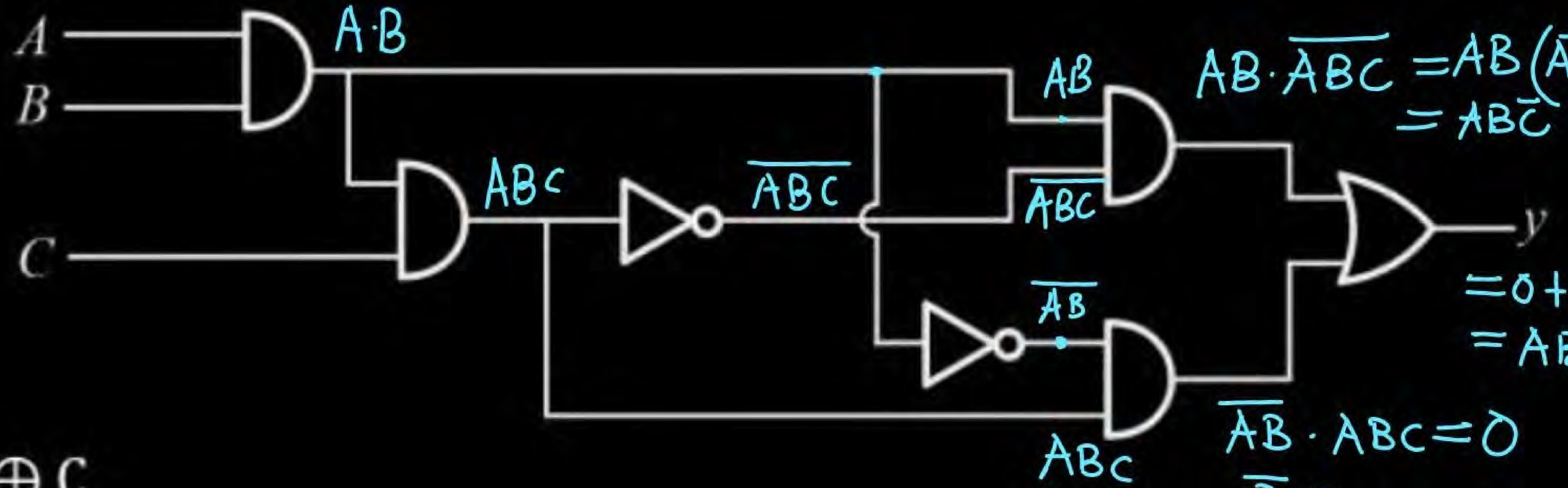
C ✓ Sum of x_1, x_2, x_3 & x_4 will be odd.

D × $\overline{x_1 x_2} + x_2 x_3 + \overline{x_3 x_4} = 1$

Question **MCQ**

A digital circuit is as given below:

Output y is



$AB\cdot\overline{ABC} = AB(\bar{A}+\bar{B}+\bar{C})$
$= AB\bar{C}$

$= 0 + AB\bar{C}$
$= AB\bar{C}$

$\overline{AB}\cdot ABC = 0$
$\bar{P}\cdot PC$
$P = AB$

A $AB\bar{C}$

B $A \oplus B \oplus C$

C $A \odot B \oplus C$

D $\overline{A+B+C}$

Question MCQ

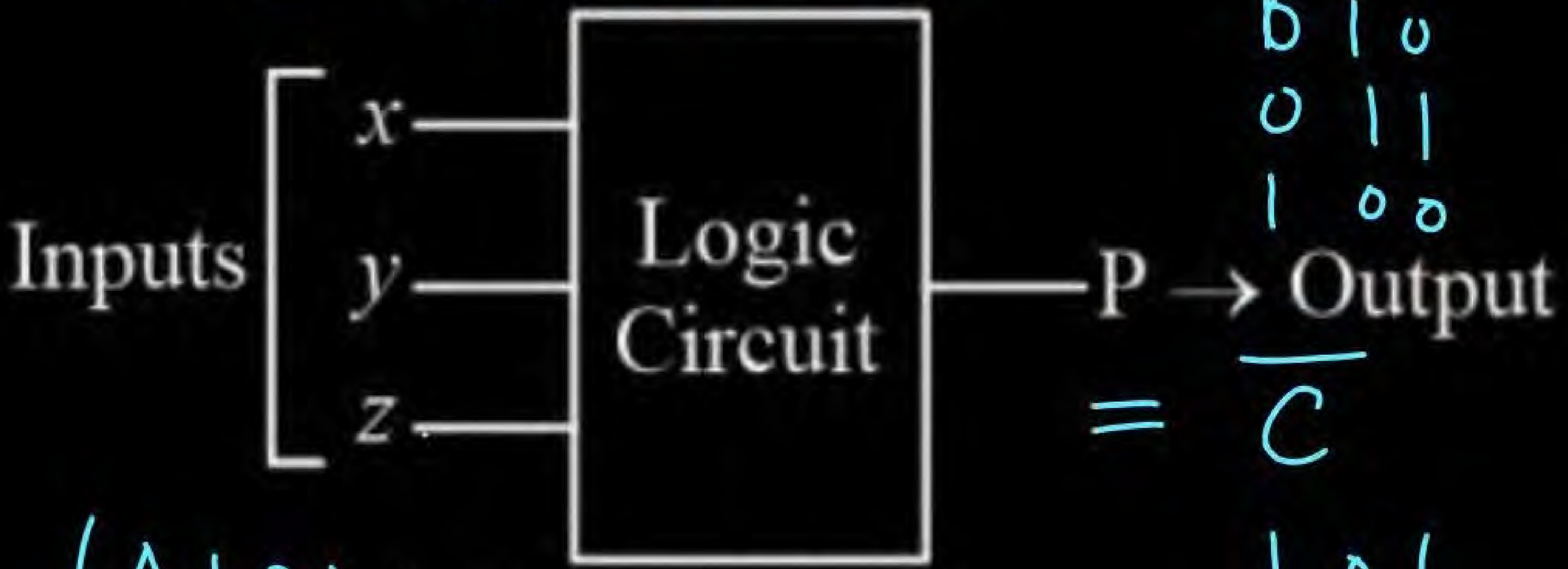
$P = \overline{Carry} = \overline{xy + yz + zx}$ $\quad Carry = \Sigma(3,5,6,7)$

A circuit is designed as shown: $= (\bar{x}+\bar{y})(\bar{y}+\bar{z})(\bar{z}+\bar{x})$ $\quad P = \Sigma(0,1,2,4)$

Output P is 1 when majority of input lines are '0'.

Output P is

$= \bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}$



$= \overline{C}$

	Carry	P
000	0	1
001	0	1
010	0	1
011	1	0
100	0	1
101	1	0
110	1	0
111	1	0

A. $xy + yz + zx$ ✗

B. $\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}$ ✓

C. $\overline{x + y + z}$ ✗

D. $\overline{xy + yz}$ ✗

$(A+B)(B+C)(C+A)$

$AB + BC + CA = (A+B)(B+C)(C+A)$

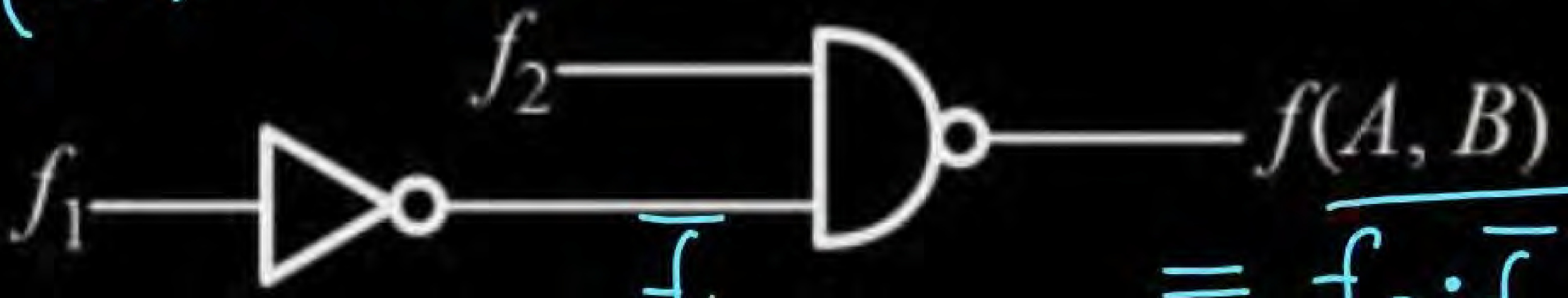
Question MCQ

Two logical functions:

$f_1(A, B) = \Sigma(1, 2)$

$f_2(A, B) = \Sigma(0, 1, 3) \longrightarrow \overline{f_2}(A, B) = \Sigma(2)$

then output f(A, B) will be



$= \overline{f_2 \cdot \overline{f_1}}$

$= \overline{f_2} + f_1$

$= \Sigma(1, 2)$

A $\Sigma(1, 2)$ ✓

B $\Sigma(0, 1, 3)$

C $\Sigma(0, 1, 2, 3)$

D $\Sigma(1)$

Question **MCQ**

A K-Map is as given below:
Its simplified solution will be

f	$\bar{B}$	B
$\bar{A}$	X	1
A	$\bar{C}$	

A $(\bar{A}+\bar{B})\bar{C}$

B $(\bar{A}+\bar{B})(\bar{A}+\bar{C})$

C $\overline{AB}+\overline{BC}$

D None of these

	$\bar{B}$	B
$\bar{A}$	X	1
A	1	

$\Rightarrow (\bar{A}+\bar{B})$

$C=0$

	$\bar{B}$	B
$\bar{A}$	X	1
A	0	

$\Rightarrow \bar{A}$

$C=1$

$= (\bar{A}+\bar{B})\bar{C} + \bar{A}\cdot C$

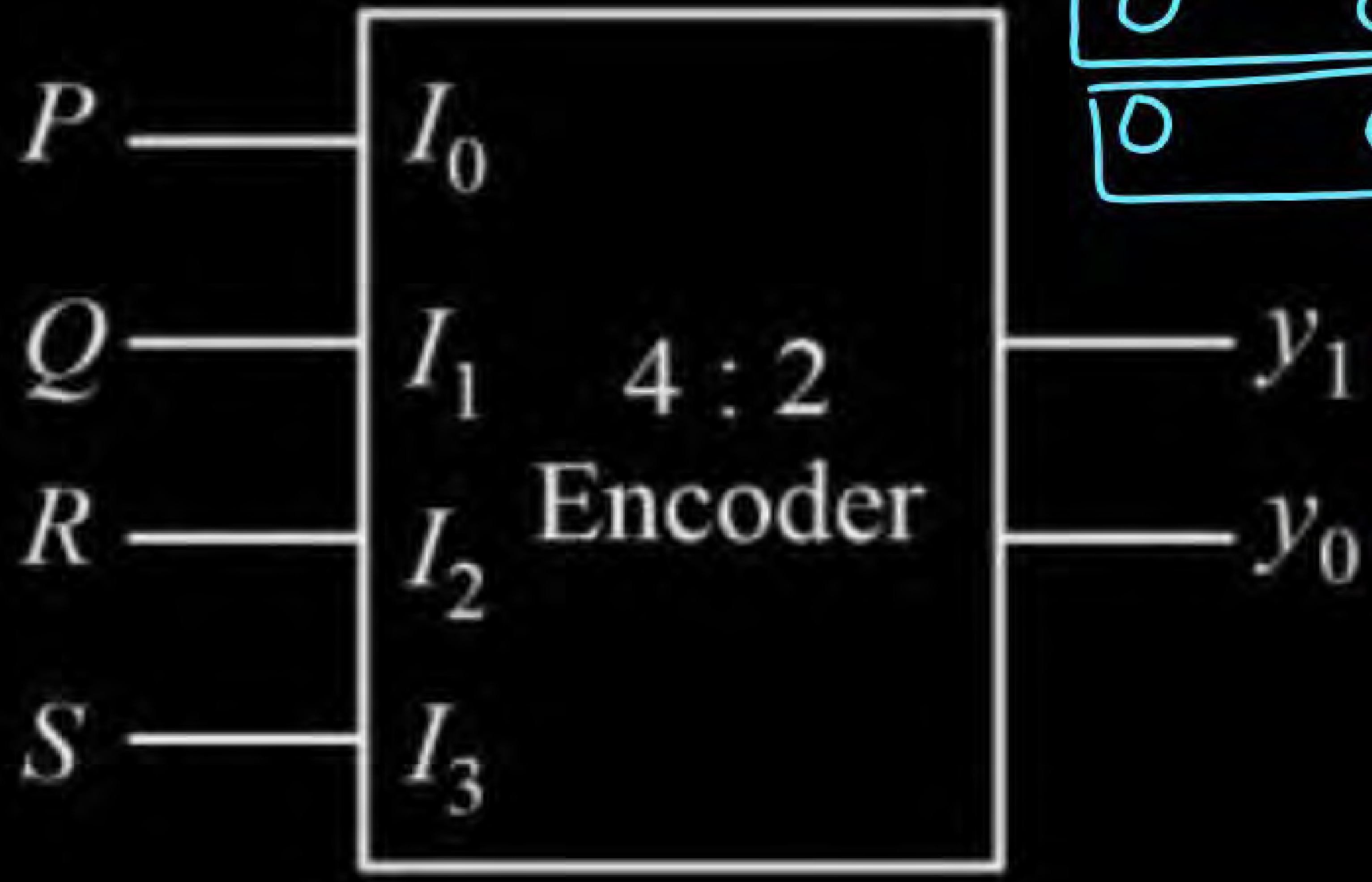
$= \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}C = \bar{A}(\bar{C}+C) + \bar{B}\bar{C} = \bar{A} + (\bar{B}\cdot\bar{C})$

$= (\bar{A}+\bar{B})\cdot(\bar{A}+\bar{C})$

Question MCQ

A 4 : 2 Encoder is as given below:
Output y_1 will be

I_0	I_1	I_2	I_3	y_1	y_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1



$y_1 = I_2 + I_3 = (R+S)$

$y_0 = I_1 + I_3 = (Q+S)$

A (P + Q)

B (Q + R)

C (R + S)

D (P + S)

Question MCQ

Which of the following is $\boxed{\text{not true?}}$

A $A + \bar{B}C = (A + \bar{B})(\bar{B} + C)$ $\xRightarrow{\text{not true}}$ $A + (\bar{B} \cdot C) = (A + \bar{B}) \cdot (A + C)$

B $\bar{A} + \bar{B}C = \overline{AB} \cdot (\bar{A} + C)$ $\xrightarrow{\text{true}}$ $\bar{A} + (\bar{B} \cdot C) = (\bar{A} + \bar{B}) \cdot (\bar{A} + C) = \overline{A \cdot B}\ (\bar{A} + C)$

C $AB + AC + \bar{B}C = AB + \bar{B}C$ true

$f\ AB + BC = (A+B)(B+C)$ not true

D $AB + BC + AC = (A + B)(B + C)(C + A)$

true

$f^{D} = (A+B) \cdot (B+C)$

Question NAT

For n-variable, the number of self dual functions are found to be 256, then the value of n is __4__.

no. of boolean functions $= 2^{2^n} = 2^{2^4} = 2^{16} = 256 \times 256$

no. of self dual functions $= 2^{2^{n-1}} = 256 = 2^8$

$2^{n-1} = 8 = 2^3$

$n-1 = 3$

$n = 4$

Question NAT

To add two 5 bit numbers, we required m H.A.s and n-OR gates then minimum value of (m +n) 13.

N → bit number ⇒

9HA
4OR

$a_4\ a_3\ a_2\ a_1\ a_0$
$b_4\ b_3\ b_2\ b_1\ b_0$
$C_4\ C_3\ C_2\ C_1$

2HA 2HA 2HA 2H.A 1H.A
1OR 1OR 1OR 1OR

Question MCQ

Two stage SOP circuit can directly be implemented

A Using NAND gates at first stage & AND gate at second stage.

B Using NAND gates only at both first & second stage.

C Using NOR gates only at both first & second stage.

D None of these

Thank you

GW Soldiers !